# Pier Review

Physics-informed mooring. Confident pier-work planning.

### THE OPERATIONAL CHALLENGE

Drafting a safe mooring and pier-work plan is complex. Bridge teams often rely on static sketches and rough estimates that cannot readily reflect changing wind, current, tide or mooring geometry.

### STATEMENT OF OBJECTIVES

Give bridge teams dynamic situational awareness to replace guesswork with calculated loads, compare configurations and prepare a shared, actionable plan for safer pier-work execution.

### HOW WE SOLVE IT

Interactive planning + automated physics

Pier Review replaces static manual sketches with a dynamic workspace that calculates bollard loads, line tensions and environmental forces as the plan changes.

- **NOAA data + NSTM 582 checks**
  Use NOAA data where available to assess geometry and loads against NSTM 582 (Mooring and Towing) referenced limits.
- **Structured mooring sequences**
  Draft line-handling sequences that support coordinated operational readiness.
- **Automatic safety sheets**
  Generate comprehensive Mooring Evaluation Safety Sheets for review and briefing.
- **Efficient tug placement**
  Configure tug positions and assistance to support precise maneuvering.
- **Intuitive graphical editing**
  Drag and drop the ship, then edit mooring connections to compare plan configurations.
- **Mapbox visual geography**
  Explore the berth and surrounding area in a rich, interactive geographic view.

APPLICATION SCREENSHOT

**Mooring layout, forces & tug placement**

APPLICATION SCREENSHOT

**Line sequence & Mooring Evaluation Safety Sheet**

### OPERATIONAL USE CASE

Before arrival or pier work, select a berth, place the ship and review wind, current and tide. Compare tug and line configurations, then brief the selected sequence and safety sheet. Reassess the plan when conditions change.

### TARGET AUDIENCE

**Primary:** bridge teams, deck officers, harbor pilots and mooring crews.

**Supporting:** tug coordinators, port operations and fleet readiness planners.

### FIELDING REQUIREMENTS

Proposed deployment configuration

Aligned with platforms already being deployed to ships, submarines and autonomous vehicles.

**Kubernetes platform**
Domino, UDS or vanilla Kubernetes to host and manage the application services.

**Dedicated GPU hardware**
Hardware acceleration for real-time physics calculation pods.

**Single sign-on**
Keycloak or enterprise SSO for identity authentication and access control.



Engineering decision support. The current demonstrator uses seed data; operational fielding requires qualified inputs and human review.